TASK 2: NETWORK SECURITY & SCANNING REPORT

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Target System: Metasploitable 2 (Linux-based Honeypot)

Target IP: 192.168.56.104

1. OBJECTIVE

The primary goal of this task was to perform a systematic security audit of the target environment. This involved discovering live hosts, mapping active services, and identifying critical vulnerabilities that could lead to system compromise.

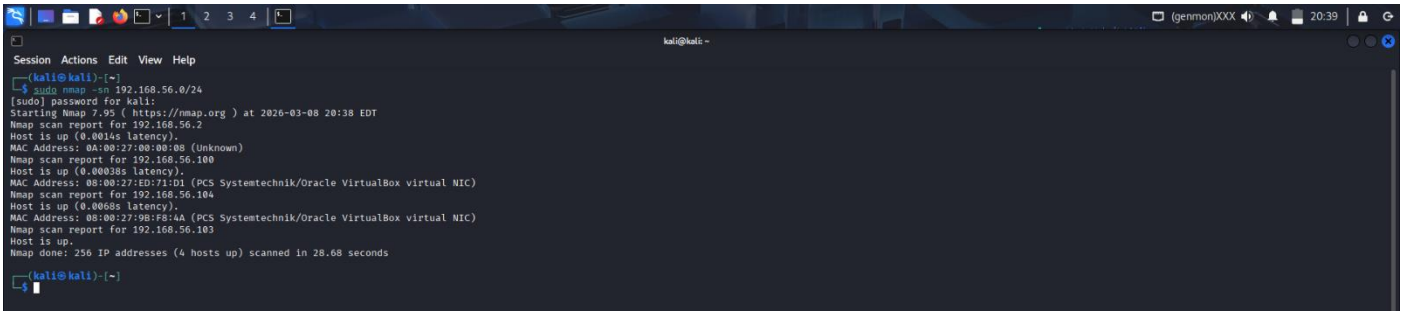
2. RECONNAISSANCE & DISCOVERY

To map the attack surface, a multi-stage scanning approach was utilized.

2.1 Network Discovery (Ping Sweep)

Command: sudo nmap -sn 192.168.56.0/24

Purpose: To identify active hosts on the host-only network without alerting security mechanisms.



Result: Identified the target at 192.168.56.104.

2.2 Detailed Service & OS Mapping

Command: sudo nmap -sS -sV -O -T4 192.168.56.104 -oN nmap_results.txt

Purpose: To perform a Stealth SYN scan, detect service versions, and identify the operating system.

Findings:

- **OS:** Linux 2.6.x
- **Open Ports:** 21 (FTP), 22 (SSH), 23 (Telnet), 80 (HTTP), 445 (Samba), 3306 (MySQL), 5432 (PostgreSQL).

3. VULNERABILITY ASSESSMENT

3.1 GVM (OpenVAS) Implementation & Troubleshooting

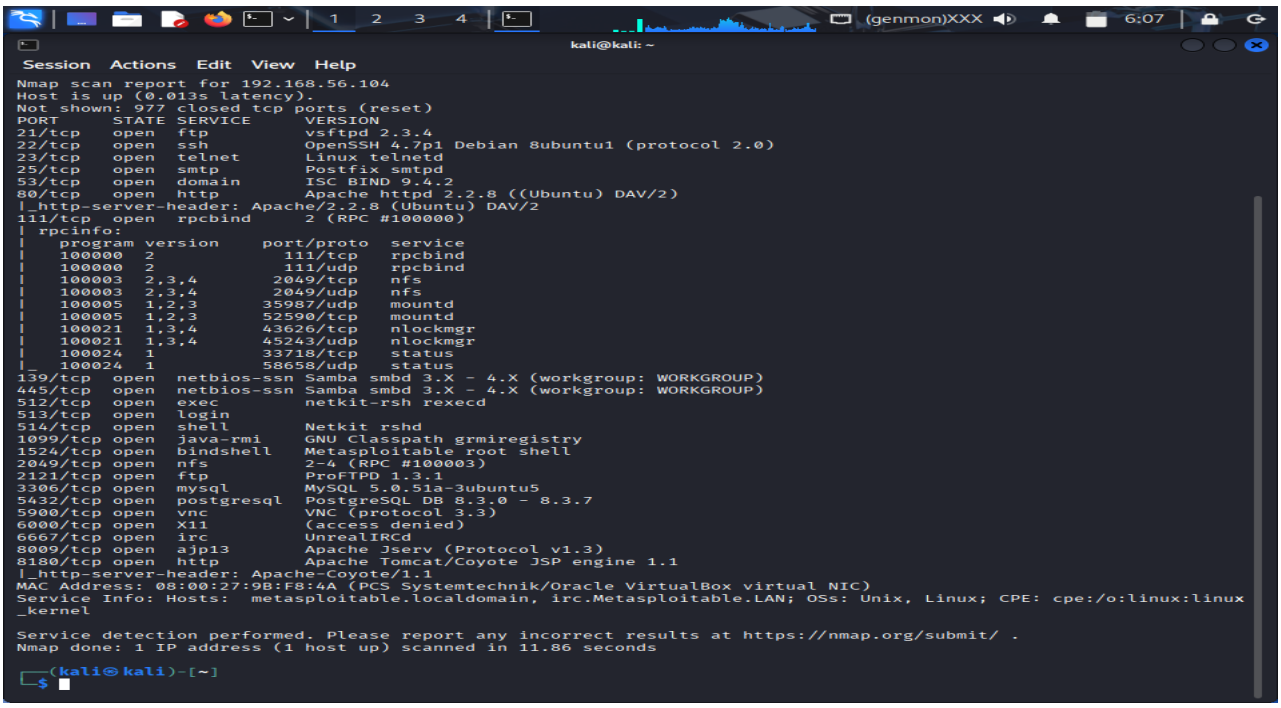
During the assessment, a full GVM (Greenbone Vulnerability Manager) environment was initialized.

- **Troubleshooting:** Encountered database synchronization issues where NVT feeds and Port Lists were pending updates. Successfully resolved GVM service locks and performed a gvm-check-setup to verify scanner readiness.

3.2 Targeted NSE Vulnerability Scan

To ensure high-fidelity results during the GVM feed synchronization, a targeted Nmap Scripting Engine (NSE) scan was executed using the vulners database.

Command: sudo nmap -sV --script=vulners 192.168.56.104



3.3 Critical Findings Table

Port	Service	CVE ID	CVSS Score	Impact
21	vsftpd 2.3.4	CVE-2011-2523	10.0 (Critical)	Malicious backdoor allowing root shell access.
445	Samba 3.0.20	CVE-2007-2447	10.0 (Critical)	Remote Command Execution via "username map script".
23	Telnet	N/A	High	Cleartext communication allows credential sniffing.

4. ANALYSIS & REMEDIATION

4.1 Vulnerability Analysis: vsftpd 2.3.4

The most critical finding is the vsftpd backdoor. This vulnerability is a classic example of a supply-chain attack where a backdoor was inserted into the source code. An attacker can gain total control of the system by simply providing a username ending in :).

4.2 Remediation Plan

1. **Patching:** Update all outdated services (Samba, Apache, vsftpd) to their latest stable versions.
2. **Deactivation:** Disable the Telnet service and migrate all remote management to SSH.
3. **Hardening:** Implement a host-based firewall (iptables) to restrict access to ports like 3306 (MySQL) and 5432 (Postgres) to local traffic only.

5. CONCLUSION

Task 2 was successfully completed. The target system was found to be in a highly insecure state with multiple CVSS 10.0 vulnerabilities. These findings provide the foundation for the exploitation phase in Task 3.

Technical Artifacts:

- **Full Nmap Output:** nmap_results.txt
- **Vulnerability Log:** task2_vulnerability_scan_report.txt